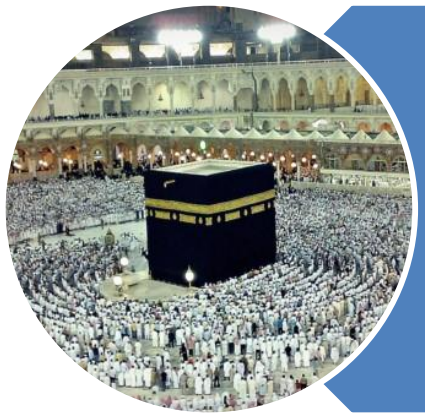
Data Structures - I

CPCS 204

رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي وَاحْلُلْ عُقْدَةً مِنْ لِسَانِي يَفْقَهُوا قَوْلِي



O my Lord! Expand for me my chest [with assurance] and ease for me my task and untie the knot from my tongue that they may understand my speech. **(Quran 20 : 25-28)**

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Course Introduction

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Course Learning Outcomes (CLOs)

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Course Learning Outcomes (CLO)

Differentiate between static and dynamic data structures. Determine the tradeoffs between different data structures and identify applications of them.	SO-b
Design and analyze more efficient algorithms for solving basic problems.	SO-j
Comprehend and trace the output of a given piece of code or algorithm.	SO-a
Demonstrate linear and binary search techniques in problem solving.	SO-a
Represent and implement linked data structures.	SO-j
Apply different operations, including search, insertion, and deletion, on linked lists.	SO-a

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Course Learning Outcomes (CLO)

Apply recursion in solving simple problems.	**SO-a**
Implement stacks using arrays and linked lists.	**SO-j**
Implement queues using arrays and linked lists.	**SO-j**
Describe and represent different tree terminologies.	**SO-a**
Apply different operations, including search, insertion, and deletion, on trees and binary search trees.	**SO-k**
Comprehend, compare, and apply sorting algorithms in problem solving.	**SO-a**

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Course Learning Outcomes (CLO)

Evaluate and Analyze the running time of small algorithms in terms of the number of operations performed	SO-b
Experiment and practice recursive sorting algorithms in problem solving	SO-b
Describe, represent, and apply hash table terminologies and operations.	SO-j
Describe, represent, and apply heap (priority queue) terminologies and operations.	SO-b
Develop small-scaled programming assignments in Java, taking space and time efficiency into account.	SO-b

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Goals
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Goals

- Improve knowledge of standard data structures and abstract data types
- Improve knowledge of standard algorithms used to solve several classical problems
- Cover some mathematical concepts that are useful for the analysis of algorithms
- Analyze the efficiency of solutions to problems
- Learn about and be comfortable with recursion

Teaching Method

- This class is NOT used to teach you JAVA
- Majority of this class is used covering new data structures, abstract data types, and algorithm analysis
- The teaching of these concepts dictate more explanation and less of a focus on "code"
  - Some code will be shown on the PowerPoint slides
    - Such as after we explain a new abstract data type
    - We'll show the code of how you would implement it
  - **However, writing of actual code will most likely never be done in class**
    - Again, that is not the purpose of this class

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Motivation
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Pre-requisites & DS

- In CPCS-202 and 203, we only cared if we found a solution to the problem at hand
  - Didn't really pay attention to the efficiency of the answer
- For this class:
  - We learn standard ways to solve problems
  - And how to analyze the efficiency of those solutions
  - Finally, we simply expand upon our knowledge of our use of the JAVA programming language

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Example Problem

- We will now go over two solutions to a problem
  - The first is a straightforward solution that a CPCS-203 student should be able to come up with
    - **Doesn't care about efficiency**
  - The second solution is one that a CPCS-204 student should be able to come up with after some thought
    - **Cares about efficiency**
- Hopefully this **example** will **illustrate** part of the **goal** of this course

Max Number of 1's

- You are given an n x n integer array
  - Say, for example, a 100x100 sized array
- Each row is filled with several 1's followed by all 0's
  - Example:
    - Row 1 may have 38 1's followed by 62 0's
    - Row 2 may have 73 1's followed by 27 0's
    - Row 3 may have 12 1's followed by 82 0's
    - You get the idea
- The goal of the problem is to identify the row that has the maximum number of 1's.

Max Number of 1's (Basic Solution)

- Straightforward CPCS-203 style solution:
  - Make a variable called MaxOnes and set equal to 0
  - For each row do the following:
    - Start from the beginning of the row on the left side
    - Scan left to right, counting the number of 1's until the first zero is encountered
    - If the number of 1's is greater than the value stored in MaxOnes, update MaxOnes with the number of 1's seen on this row
- Clearly, this works
- But let's see how long this algorithm will take

Max Number of 1's (Example 01)

1	1	0	0	0	0
0	0	0	0	0	0
1	1	1	0	0	0
1	1	1	0	0	0
1	1	1	1	1	0
1	1	1	1	0	0

Max Number of 1's (Example 01)

1	1	0	0	0	0
0	0	0	0	0	0
1	1	1	0	0	0
1	1	1	0	0	0
1	1	1	1	1	0
1	1	1	1	0	0

- Rows 6 & Columns 6
- # of Comparison 23
- Max. Number of comparisons 6 x 6 = 36

Max Number of 1's (Example 02)

0	0	0	0	0	0	0	0
1	1	0	0	0	0	0	0
1	0	0	0	0	0	0	0
1	1	1	0	0	0	0	0
1	1	1	1	0	0	0	0
1	1	1	1	1	1	0	0
1	1	1	1	1	1	1	0

- Rows 7 & Columns 8
- # of comparisons 30
- Max. Number of comparisons 7 x 8 = 56

Max Number of 1's (Analysis)

- Analysis of Straightforward Solution:
  - Basically we iterate through each square that contains a 1, as well as the first 0 in each row
  - If all cells were 0, we would only "visit" one cell per row, resulting in n visited cells
  - However, if all cells were 1's, we would "visit" all of the cells (n^2 total)
    - So in the worst case, the number of simple steps the algorithm takes would be approximately n^2
  - This makes the running time of this algorithm $O(n^2)$
    - The meaning of this Big-O will be discussed later in the semester

Max Number of 1's (Optimal Solution)

- More Efficient CPCS-204 style algorithm:

1. Initialize the current row and current column to 0
2. While the current row is less than n (or before the last row)
   a. While the cell at the current row and column is 1
      - Increment the current column
   b. Increment the current row
3. The current column index represents the maximum number of 1's seen
4. Now let's trace through a couple of examples

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Max Number of 1's (Example 01)

	0	1	2	3	4	5
0	1	1	0	0	0	0
1	0	0	0	0	0	0
2	1	1	1	0	0	0
3	1	1	1	0	0	0
4	1	1	1	1	1	0
5	1	1	1	1	0	0

- Rows 6 & Columns 6
- # of comparisons 11 (previously 23)
- Max. Number of comparisons 6 + 6 = ~~12~~ 11

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Max Number of 1's (Example 02)

0	0	0	0	0	0	0	0
1	1	0	0	0	0	0	0
1	0	0	0	0	0	0	0
1	1	1	0	0	0	0	0
1	1	1	1	0	0	0	0
1	1	1	1	1	1	0	0
1	1	1	1	1	1	1	0

- Rows 7 & Columns 8
- # of comparisons 14 (previously 30)
- Max. Number of comparisons 7 + 8 = 15

Max Number of 1's (Analysis)

- Analysis of Better Solution:
  - How many steps will this algorithm take, in terms of n?
    - Each "step" taken by the algorithm either goes to the right or down in the table.
    - There are a maximum of n-1 steps to the right
    - And a maximum of n-1 steps down that could be taken
    - Thus the maximum number of "steps" that can be done during this algorithm is approximately **2n – 1**
      - And this is the worst case
    - So the running time of this algorithm is **O(n)**
      - An improvement of the previous algorithm
      - Input size of 100 for n
      - n^2 would be 10,000 steps and 2n would be 200 steps

Implementing an Algorithm in JAVA

- In this class, you will have an opportunity to improve upon your ability to write programs that implement an algorithm you have learned
- You must know the syntax of JAVA in order to properly and effective do this
- There's no set way to create code to implement an algorithm
- We have many set algorithms and data structures that you will study
- Mostly, however, you will simply have to apply the data structures and algorithms shown in class fairly directly to solve the given problems

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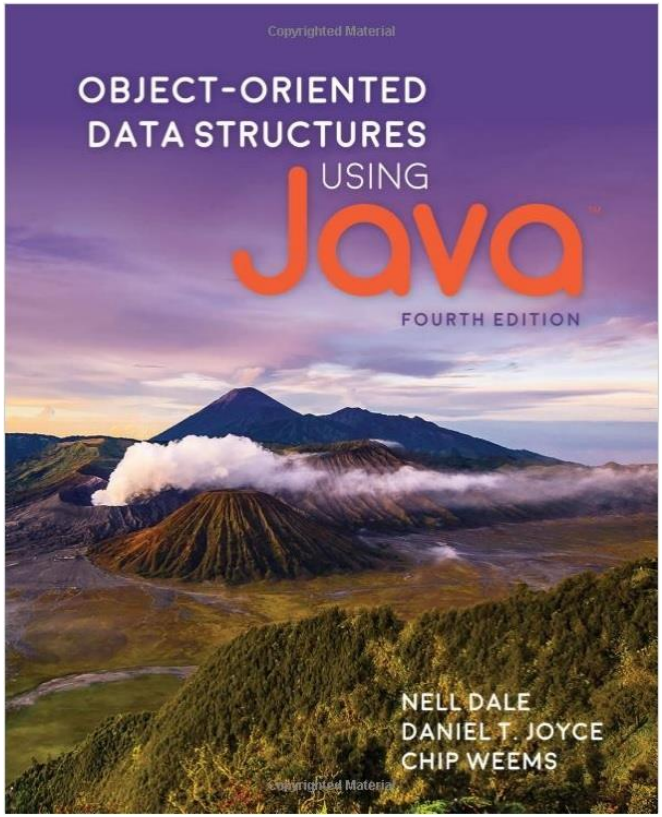
Text Book

"Object-Oriented Data Structures Using Java",

Nell Dale, Jones & Bartlett Learning; 4th edition (September 12, 2016).

ISBN-10: 1284089096.

ISBN-13: 978-1284089097.



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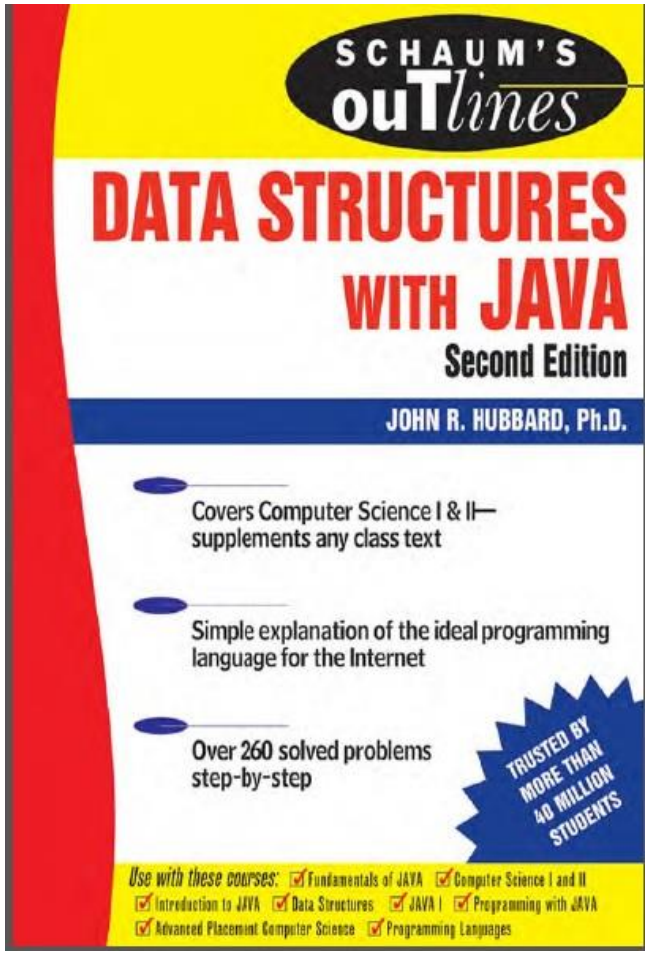
Reference Book

"Schaum's Outlines, Data Structures with Java",

John R. Hubbard; McGraw-Hill; 2nd edition (2007). ISBN-07-147698-9.



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General Policies

- Attendance must be 75% or more
- All written exams will be cumulative
- Quizzes will be on blackboard, No Makeup
- Plagiarism / cheating will result in 0% marks
- Pay regular visit to Blackboard
- Need help for Assignments; contact lab instructor
- **No scale up / bonus for final grade**

Assessments

Assessment	Grading Percentage
Quizzes	05%
Lab Work	15%
Assignments (3 * 5% each)	15%
Exam 1	15%
Exam 2	20%
Final Exam	30%

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"No Bonus" marks for this course anymore

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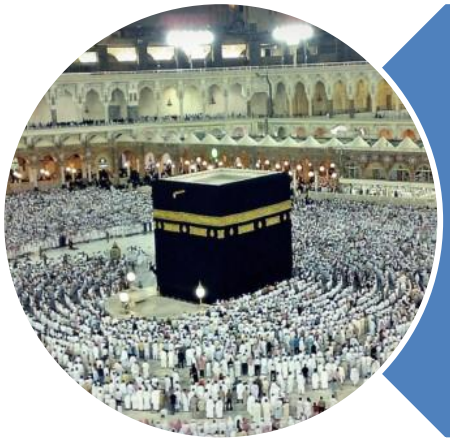

Allah (Alone) is Sufficient for us, and He is the Best Disposer of affairs (for us).(Quran 3 : 173)

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